CONNECTOR PINOUT & ELECTRODE GUIDE · SKU CQA240229P1

Electrodes, Header & GPIO Pins

How to connect 8 electrodes, the reference and the bias to PiEEG-8, the default 10-20 montage, and the five Raspberry Pi GPIO pins the shield uses.



8	REF	BIAS	5	2.54
Electrode channels	Ear clip · common 0 µV	Ear clip · noise rejection	GPIO signal pins used	mm header pitch

01 ELECTRODE INPUTS

Input	Count	Electrode	Purpose
Channels 1–8	8	Dry or wet Ag/AgCl, in the cap	Voltage between each electrode and the reference
Reference	1	Ear clip on one earlobe	Common 0 µV point for all 8 channels
Bias	1	Ear clip on the other earlobe	Drives the body to cancel common-mode noise such as 50/60 Hz hum

02 ELECTRODE HEADER

Square pad = pin 1 (E1). 2.54 mm pitch, single row.

E1	E2	E3	E4	E5	E6	E7	E8	Biasout	Ref
pin 1	pin 2	pin 3	pin 4	pin 5	pin 6	pin 7	pin 8	pin 9	pin 10
Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7	Channel 8	Bias ear clip	Reference ear clip

Figure 1 — Electrode header as printed on the board (top view).

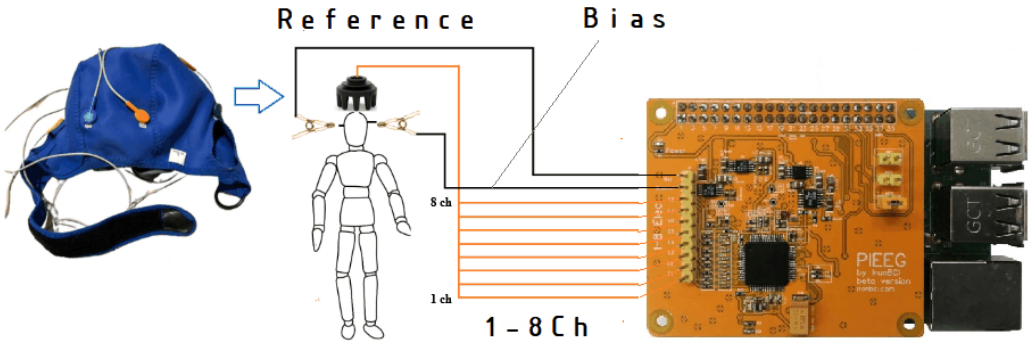
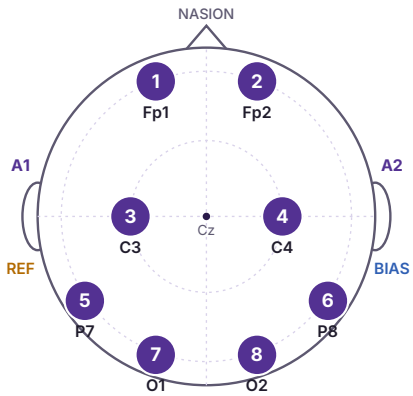


Figure 2 — Wiring of channels 1–8, reference and bias from the cap to the shield.

> **NOTE Reference and bias clips matter most.**
> A loose ear clip makes every channel noisy at once. Clean the earlobes and make sure both clips sit firmly before looking at anything else.

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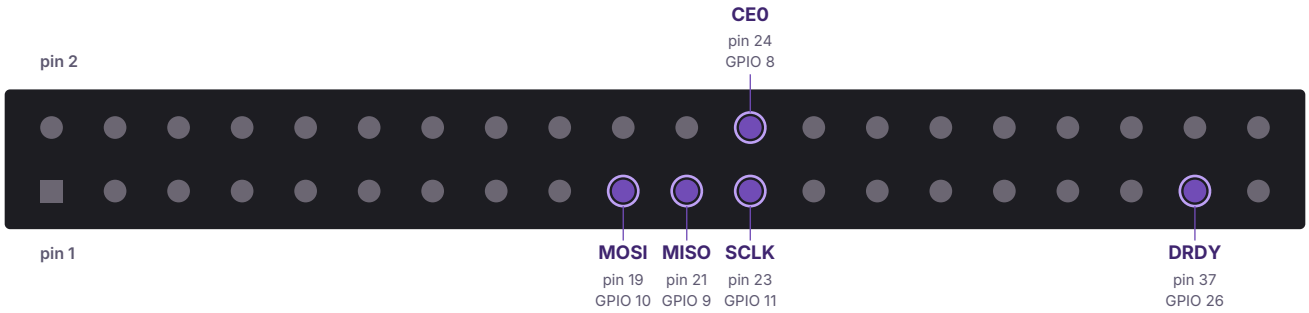
03 DEFAULT MONTAGE (10-20 SYSTEM)



Ch	Position	Area	Good for
1	Fp1	Front left	Blinks, eye movement
2	Fp2	Front right	Blinks, eye movement
3	C3	Centre left	Motor activity (right hand)
4	C4	Centre right	Motor activity (left hand)
5	P7	Back left, side	Parietal / temporal
6	P8	Back right, side	Parietal / temporal
7	O1	Back left	Alpha rhythm, vision
8	O2	Back right	Alpha rhythm, vision

This is BrainFlow's default naming for PiEEG. Any other positions work — note which channel went where.

04 RASPBERRY PI GPIO PINS



Odd pins: bottom row · even pins: top row · power and ground are also taken from the header

Header pin	GPIO	Signal	Direction
19	GPIO 10	SPI0 MOSI	Pi → ADS1299
21	GPIO 9	SPI0 MISO (DOUT)	ADS1299 → Pi
23	GPIO 11	SPI0 SCLK	Pi → ADS1299
24	GPIO 8	SPI0 CE0 (chip select)	Pi → ADS1299
37	GPIO 26	DRDY (data ready, active low)	ADS1299 → Pi

The shield also takes power and ground from the header. Other GPIO pins are free for your own buttons, LEDs or triggers, if you can reach them with the shield mounted.

05 FITTING THE CAP

Step	Action
1	Measure the head and pick the right cap size.
2	Find **Cz** (halfway nose–back of head, halfway between the ears) and align the cap.
3	**Dry electrodes:** part the hair so each electrode touches the scalp. **Wet:** work gel through the hair with the blunt syringe.
4	Attach the ear clips to clean earlobes: REF on one, BIAS on the other.
5	Ask the person to sit still and relax the face and jaw.

06 RELATED DOCUMENTS

PE8-DS-001 Product Datasheet	**PE8-MD-002** Mechanical & Dimensional Drawing	**PE8-PCB-003** PCB & Manufacturing Specification
PE8-QSG-005 Quick Start & Safety Guide	**PE8-SW-006** Data Interface & Software Guide	**PE8-ENC-007** 3D-Printed Enclosure Guide

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